

MODEL EVALUATION REPORT

Customer Churn Prediction – Week 3 Machine Learning Project

1. Introduction

This report evaluates four classification models developed for the customer churn prediction problem: Logistic Regression, Decision Tree, Random Forest, and XGBoost. The objective is to determine which model provides the most useful predictive performance for identifying customers who are likely to churn.

2. Evaluation Metrics

The models were evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Accuracy measures the proportion of all predictions that are correct. Precision measures the proportion of customers predicted to churn who actually churned. Recall measures the proportion of actual churners correctly identified. F1-score provides a balance between Precision and Recall. ROC-AUC measures the model's ability to distinguish between churn and non-churn customers across classification thresholds.

Because the target distribution contains 5,174 non-churn customers (73.46%) and 1,869 churn customers (26.54%), Accuracy was not treated as the only criterion. Recall, F1-score, and ROC-AUC were given particular consideration for the churn-retention objective.

3. Model Performance Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
XGBoost	0.7424	0.5092	0.8102	0.6254	0.8430
Logistic Regression	0.7381	0.5043	0.7834	0.6136	0.8417
Random Forest	0.7821	0.6184	0.4679	0.5327	0.8222
Decision Tree	0.7360	0.5026	0.5080	0.5053	0.6637

4. Model-by-Model Evaluation

XGBoost achieved an Accuracy of 0.7424, Precision of 0.5092, Recall of 0.8102, F1-score of 0.6254, and ROC-AUC of 0.8430. It achieved the highest Recall, F1-score, and ROC-AUC among the four models. Logistic Regression achieved 0.7381 Accuracy, 0.5043 Precision, 0.7834 Recall, 0.6136 F1-score, and 0.8417 ROC-AUC, making it a competitive and simpler baseline. Random Forest achieved the highest Accuracy (0.7821) and Precision (0.6184), but its Recall was 0.4679 and F1-score 0.5327. Decision Tree achieved 0.7360 Accuracy, 0.5026 Precision, 0.5080 Recall, 0.5053 F1-score, and 0.6637 ROC-AUC, giving it the weakest overall discrimination.

5. Model Comparison

XGBoost ranked first when considering the metrics most relevant to the stated churn objective. Its ROC-AUC of 0.8430 was the highest, followed closely by Logistic Regression at 0.8417. XGBoost also had the highest Recall (0.8102) and F1-score (0.6254). Random Forest led in Accuracy and Precision, but its lower Recall means it identified substantially fewer actual churners. Decision Tree was the weakest overall model, particularly in ROC-AUC.

6. Final Model Recommendation

XGBoost is recommended as the preferred model for this customer churn prediction task. It provides the strongest overall balance of discrimination and churn detection, with the highest ROC-AUC, Recall, and F1-score. The recommendation is based on the business trade-off represented by the evaluation results, not on a single metric.

7. Business Implications

The selected model can be used to identify customers with elevated churn risk and prioritize retention interventions. The business should consider predicted risk together with customer value and the cost of retention actions when defining intervention thresholds. Performance should be monitored after deployment and validated on new customer data.

8. Conclusion

The four models showed different performance profiles. XGBoost provided the best overall performance for the churn-prediction objective, while Random Forest led in Accuracy and Precision. Logistic Regression offered a competitive baseline, and Decision Tree had the weakest overall discrimination. Based on the evaluation results, XGBoost was selected as the preferred model for supporting targeted customer-retention activities.